

Imbalanced Data - Health Insurance Dataset

Francisco da Ana (up202108762)
(Dated: December 28, 2024)

This report examines machine learning techniques designed to address the challenges of imbalanced datasets, where one class is significantly underrepresented. Such imbalance can severely affect model performance, resulting in biased predictions and poor generalization. To deal with this issue, some approaches will be explored - such as data resampling and specialized evaluation metrics. By applying these strategies, machine learning models can achieve better performance, even in highly imbalanced scenarios.

I. INTRODUCTION

In previous assignments, I explored various aspects of "Data-Centric & AI Data Profiling" and "Data Complexity & Meta-Learning", using a dataset designed to predict whether customers have **health insurance (binary classification problem)**. This dataset presented a **significant imbalance in the target variable**, which posed challenges for building unbiased and accurate models.

For this assignment, I have chosen to continue working with the same dataset to extend the case study and address the challenges posed by class imbalance. This report presents techniques to deal with this problem and develop a model capable of accurately predicting the target variable without introducing bias.

Before addressing the solutions, I will perform a brief data profiling step to identify and resolve key issues in the dataset, establishing a reliable foundation for the analysis.

II. HEALTH INSURANCE DATASET

This analysis continues the exploration of the Health Insurance dataset previously examined in previous assignments. The dataset comprises about 72,000 instances with 15 features, where the binary target variable *health_ins* indicates whether an individual has health insurance or not.

III. DATA PROFILING SUMMARY

The primary issue identified in the dataset is the high imbalance ratio of the target variable, which will be addressed in a further section. Before moving to that, the following preprocessing steps were carried out based on data profiling results:

Outliers: Feature *age* was truncated to the range 21–99 (years old) and scaled using min-max scaling. Features such as *gas_usage* and *income*, which were highly right-skewed, were transformed using logarithmic normalization.

Missing Values: Missing values in feature *is_employed* represented individuals who are not in

the workforce. Given the high imbalance in this attribute, they were treated as "not employed". Missing data, found in the same rows and features, was not the focus of this assignment. Thus, rows containing missing values (approximately 2%) were removed.

Feature Selection: Columns representing unique identifiers and redundant information were excluded from the dataset.

IV. IMPACT OF IMBALANCED DATA

Before applying techniques to address the imbalance issue, I conducted a brief analysis to assess how this imbalance affected the performance of machine learning algorithms. For this analysis, I chose two baseline algorithms — Decision Tree and KNN — due to their simplicity and good computational efficiency. The dataset was divided into two partitions: training (70%) and validation (30%).

Both models appeared to achieve impressive performance metrics: accuracy 91% and F1 score 95%. But do these results indicate that the models are reliable and capable of generalizing to real-world scenarios? To explore this further, I examined the confusion matrices and the results were impressive.

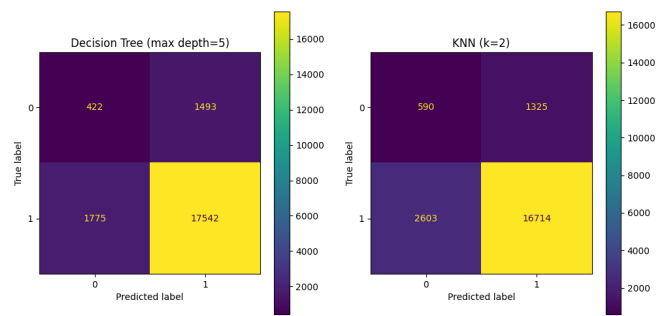


FIG. 1: Confusion matrices of Decision Tree and KNN trained on imbalanced dataset

The confusion matrices revealed a critical issue: neither model was able to accurately detect the minority class (negative cases). With an **imbalance ratio of 10** (positives/negatives), these performance metrics were highly misleading, as both models were biased toward the majority class. While accuracy and F1 score suggested

strong performance, they failed to provide meaningful insights, leading to **overoptimistic conclusions**.

This misleading result occurred because the validation partition, as did the training data, reflected the **same class distribution**. As a result, even a biased model tested on an imbalanced partition can show high accuracy and F1 score. To deeper understand the performance of the models, I calculated **specificity** (the percentage of negative cases correctly classified), and the results were concerning: the decision tree achieved 20% specificity, while KNN reached 30%. These values clearly highlight the severity of the problem, as the **models were poorly capable of detecting the minority class**.

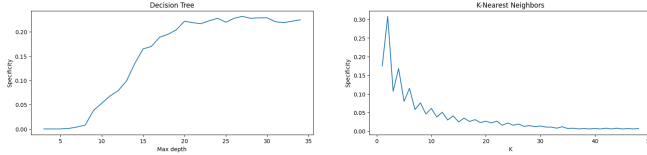


FIG. 2: Impact of hyperparameters (maximum depth and number of neighbors) in specificity

I also explored how hyperparameters, specifically maximum depth for the Decision Tree and the number of neighbors for KNN, affect specificity. The results showed that a **deeper Decision Tree** improved the model's ability to detect the minority class, as it allowed the model to capture more complex patterns in the data. Similarly, for KNN, a **lower number of neighbors** favored the detection of the minority class. With fewer neighbors, the model focused more on local patterns, which helped it better identify the minority instances. These findings highlight the importance of fine-tuning hyperparameters to improve the performance of models on imbalanced datasets.

V. DEALING WITH IMBALANCE

Once the problem was identified, I decided to apply **rebalancing techniques** to address the class imbalance. SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic examples of the minority class by interpolating between existing instances. Given the high complexity of the classification problem, I selected several SMOTE variants and tested them to choose the best one. The rebalancing techniques tested included "classical" SMOTE, random oversampling, Borderline-SMOTE, ADASYN, SVM-SMOTE, SMOTE-ENN, and SMOTE-Tomek links.

To evaluate the quality of the rebalanced datasets, I considered the **feature-based data complexity measures** studied in the previous assignment: F1 (class separability), F2 (feature overlap), F3 (best single feature), and F4 (combined feature effectiveness). The goal was to find a balance between achieving a **high class separability**

(F1) and **minimizing feature overlap** (F2) to ensure the dataset is well-suited for accurate, unbiased model training.

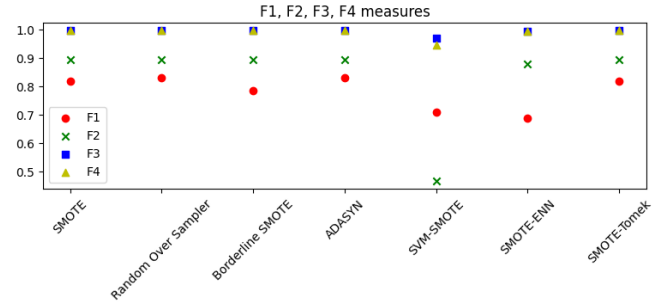


FIG. 3: Comparison of feature-based measures related to rebalanced datasets

SVM-SMOTE was chosen as the best technique, because even with a slightly lower F1 measure, it has a significant lower F2 measure.

VI. RESULTS AND CONCLUSION

After selecting a rebalanced dataset, I tested the previous models again, and they became much more capable of capturing negative cases (the minority class). **Specificity increased to around 70%**, indicating a significant improvement in the detection of the minority class. However, the traditional performance metrics, such as accuracy and F1-score, decreased slightly. This was expected, as the validation set no longer followed the same distribution as the training set (imbalanced vs. balanced). Despite the decrease in these metrics, the improvement in specificity demonstrates that the **rebalancing techniques effectively addressed the class imbalance issue**.

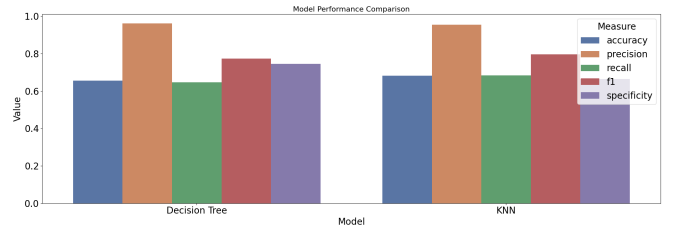


FIG. 4: Performance of Decision Tree and KNN in rebalanced dataset

REFERENCES

- [1] Matthew Stewart. *Imbalanced Datasets: Complete Guide to Classification*. 2020. URL: <https://resources.experfy.com/ai-ml/imbalanced-datasets-guide-classification/>.

- [2] Cornelius Yudha Wijaya. *7 SMOTE Variations for Oversampling*. 2023. URL: <https://www.kdnuggets.com/2023/01/7-smote-variations-oversampling.html>.
- [3] O. Olawale Awe. *Handling Data Imbalance in Machine Learning*. 2024. URL: <https://isi-web.org/sites/default/files/2024-02/Handling-Data-Imbalance-in-Machine-Learning.pdf>.
- [4] The imbalanced-learn developers. *imbalanced-learn documentation*. 2024. URL: <https://imbalanced-learn.org/stable/>.